

GEME: Self-Organizing Concept Hierarchies through Recursive Co-occurrence

A computational model of emergent symbol-grounding in formal systems

Abstract

We present GEME (GodEi-Memory Economy), a minimal computational model demonstrating that concept hierarchies can emerge from a single primitive: merging structurally similar formulas in a competitive memory economy. Operating with no gradient descent, no pre-defined targets, and no external ontology, GEME receives discrete geometric assertions and spontaneously organizes them into a multi-level structure: (S0) raw input frames, (L1) pairwise association links via temporal co-occurrence, (L2) proof-path chains bridging heterogeneous concept types, and (L3+) meta-theoretic recursion. After 5,000 unsupervised steps, the system converges to a stable 15-frame memory containing three core concepts (triangle, segment, angle), a complete 6-link association graph, and chained inference paths -- all without any external supervision or pre-coded geometric knowledge.

Architecture

The standard representation vector (19 dimensions) encodes each formula by counting symbol frequencies across the alphabet: point labels (A..F), geometric functions (seg, angle, triangle), relations (equiv, conj, impl), and delimiters.

Input formulas enter sequentially. Each formula is compared against all existing memory frames using Levenshtein distance on their structural signatures. If the distance is below threshold (0.25), the new formula merges into the nearest frame and its weight increments by 1.0. Otherwise it creates a new frame.

When memory stress (utilization times inefficiency) exceeds 0.6, the system triggers induction: it retains only the top half of frames by weight, allowing infrequent patterns to decay while frequent ones consolidate.

Simultaneously, a sliding window of the last 60 signatures tracks co-occurrence. Pairs appearing together in more than 25% of window positions create an association frame. When two association frames share a base signature element (e.g., both contain the "seg" symbol), the system creates a chain frame linking them. This process recurses to any depth, building a layered structure.

Hierarchy summary:
Level 0 - MAIN frames: highest-weight input patterns (wt 1200-1300)
Level 1 - Association links: co-occurrence pairs (wt 32)
Level 2 - Chain paths: shared-element bridges (wt 15)
Level 3+ - Meta-recursion: unlimited depth via exact-match dedup

Key Mechanism

Memory economy: Each input formula is encoded as a frequency vector over a 19-symbol alphabet (seg, angle, triangle, equiv, conj, impl, and point labels). Similar vectors merge into a single frame, incrementing weight. When stress (utilization times inefficiency) exceeds threshold, induction evicts low-weight frames. A 5,000-step input stream is spontaneously compressed to about 15 stable frames.

Recursive co-occurrence: A sliding window (size 60) tracks which formula signatures appear together. Pairs exceeding 25 percent co-occurrence create association frames. When associations share a base element (for example, both contain "seg"), a chain frame is created. Exact-match deduplication prevents infinite recursion while allowing unlimited self-reference depth.

Emergent hierarchy: The weight gap between levels (40:1 from Level 0 to Level 1, 2:1 from Level 1 to Level 2) is not engineered -- it emerges from the asymmetry between raw frequency, correlation, and second-order structure. This mirrors Hofstadter's stratified symbol systems.

Experimental Results

Metric	Arithmetic Stage (PA)	Geometry Stage (GO)
Axioms required	7 (Peano)	0 (random stream)
Input steps	500	5,000
Stable frames	2 to 4	15 of 15 capacity used
Core concepts	commutative laws	triangle, segment, angle

Association	external S4 signal	endogenous L1 co-occurrence
Inference paths	none	L2 chain frames (3 to 4)
Code size	178 lines	200 lines
Dependencies	none	none

After 5,000 unsupervised steps from random geometric assertions: triangle (weight 1,289), segment (weight 1,287), angle (weight 1,248) as core Level 0 concepts; a complete 6-link association graph at Level 1 (weight 32); and 3 to 4 inference-chain frames at Level 2 (weight 15). The concept "parallel" was correctly demoted to derived status (appearing only at Level 1, never as a main frame). A generalization test confirmed that training on only SAS and SSS theorem patterns enabled recognition of ASA upon first exposure (weight 101 independent frame) -- the system learns meta-patterns rather than memorizing specific templates.

Relation to Existing Frameworks

GEME is orthogonal to mainstream approaches. Unlike the Free Energy Principle (Friston, 2024) which minimizes prediction error via Bayesian inference, GEME minimizes encoding stress via competitive memory -- no generative model is required. Unlike usage-based grammar induction (MIT Computational Linguistics, 2024) which batch-processes n-gram statistics, GEME operates online with no separate pattern-extraction phase. Unlike neuro-symbolic systems (DeepMind), GEME requires no neural component -- a single syntactic distance metric suffices for both perception and reasoning. Unlike Godel machines (Schmidhuber, 2003) which search proof spaces for self-modification, GEME uses no proof search -- self-reference emerges naturally from co-occurrence tracking. GEME provides the first concrete computational implementation of Hofstadter's strange-loop hypothesis (Goedel, Escher, Bach, 1979): recursive self-reference in a competitive memory substrate spontaneously generates hierarchical symbol systems.

Conclusion

GEME demonstrates that concept formation -- primitive grounding, association discovery, inference-path construction, and meta-theoretic recursion -- can emerge from a single operation (structural merge) and a single mechanism (co-occurrence tracking) without external objectives or architectural layering. The system's 200-line implementation contrasts sharply with million-parameter models in contemporary AI, suggesting that symbol emergence may be simpler than assumed -- if one adopts memory economy as the primitive. We offer this work as both a technical contribution and a tribute to the ideas that inspired it.

Inspired by Hofstadter's Goedel, Escher, Bach (1979). Personal research project.